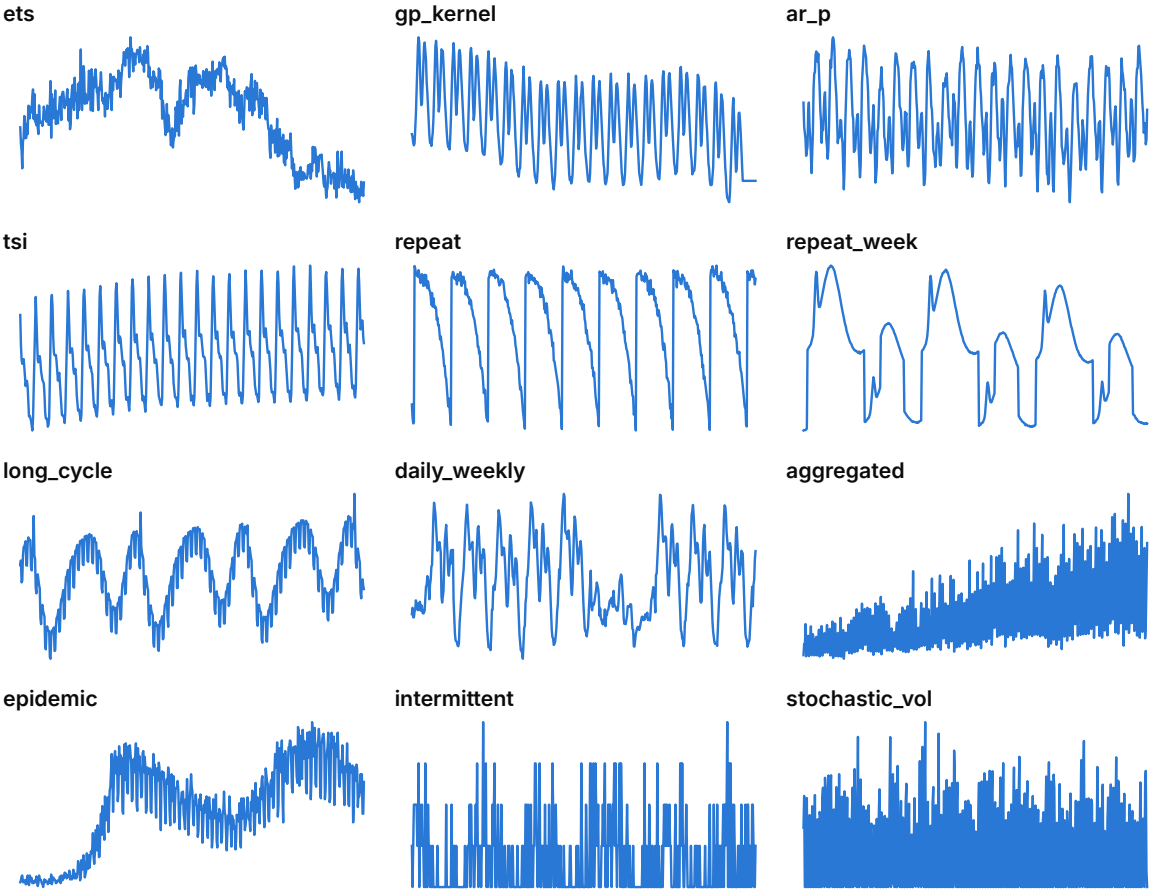One random draw per family (512 steps each, all layers applied)